AUTOMORPHISMS RESTRICT TO NORMAL SUBEXTENSIONS, AND MAPS INTO A NORMAL EXTENSION EXTEND TO AUTOMORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The xymatrix diagram (source lines 2181-2185) is part of the counted proof span and renders with the chapter preamble's xy package loading.

Editorial note. The label lemma-lift-maps names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: theorem-existence-algebraic-closure (delivered card L48331), lemma-map-into-algebraic-closure (delivered card L48339: K-embeddings extend along algebraic extensions), lemma-characterize-normal (normality as the equal-image criterion for F-embeddings into an algebraic closure)

Source reference. Original reference 09HQ: sources/stacks-fields/pinned-original-source.tex, source lines 2096–2096. Prerequisite pointer invoked at 2179; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GU: sources/stacks-fields/pinned-original-source.tex, source lines 1240–1240. Prerequisite pointer invoked at 2184; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GT: sources/stacks-fields/pinned-original-source.tex, source lines 1184–1184. Prerequisite pointer invoked at 2175; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $L/M/K$ be a tower of algebraic extensions.*

(1) *If M/K is normal, then any automorphism τ of L/K induces an automorphism $\tau|_M : M \to M$.*
(2) *If L/K is normal, then any K-algebra map $\sigma : M \to L$ extends to an automorphism of L.*

Proof. Choose an algebraic closure $\overline{L}$ of L (Theorem 09GT).

Let τ be as in (1). Then $\tau(M) = M$ as subfields of $\overline{L}$ by Lemma 09HQ and hence $\tau|_M : M \to M$ is an automorphism.

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Let $\sigma : M \to L$ be as in (2). By Lemma 09GU we can extend σ to a map $\tau : L \to \overline{L}$, i.e., such that

$$\begin{array}{ccc} L & \xrightarrow{\tau} & \overline{L} \\ \uparrow & \nearrow_{\sigma} & \uparrow \\ M & \longleftarrow & K \end{array}$$

is commutative. By Lemma 09HQ we see that $\tau(L) = L$. Hence $\tau : L \to L$ is an automorphism which extends σ. $\square$